Shubham Vishwakarma

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Portfolio

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Education

- **SPPU - AISSMS IOIT** — Pune
  B.E. in Electronics and Telecommunication; CGPA: 8.88/10 — *April 2023*

Experience

- **TCS (Microsoft Partner Staffing 360)** — Pune
  Full Stack Developer — *March 2024 – Present*
  - **AI Adoption Leadership**: Led AI adoption initiatives across the project by designing and integrating AI agents to automate feature development workflows, improving developer productivity and reducing manual effort.
  - **Spec Kit Implementation**: Owned and implemented Spec Kit architecture to standardize feature development, enabling faster delivery and consistent engineering practices across the team.
  - **AI Agent Training & Optimization**: Designed and trained AI agents using structured prompts and Figma-based UI inputs to generate accurate, production-ready code aligned with business requirements.
  - **UI-to-Code Automation**: Leveraged Figma design screens to train AI systems for UI generation, significantly reducing frontend development time and ensuring pixel-perfect implementation.
  - **Full Stack Engineering**: Developed scalable web applications using React.js and .NET (C#), contributing to Microsoft partner onboarding systems.
  - **API & System Design**: Built and optimized RESTful APIs with focus on performance, maintainability, and seamless frontend-backend integration.
  - **Cloud & DevOps**: Deployed and managed services on Azure (App Services, Functions, Key Vault) and contributed to CI/CD pipelines using Azure DevOps.
  - **Cross-functional Collaboration**: Worked closely with designers, product owners, and backend teams to translate business requirements into scalable technical solutions.
  - **Code Quality & Mentorship**: Actively contributed to code reviews, guided team members on AI-driven development practices, and improved overall code quality standards.

- **LTIMindtree** — Remote
  Graduate Engineering Trainee — *Feb 2023 – May 2023*
  - **Training**: Completed training in Java programming and SQL database management with distinction.
  - **Performance**: Achieved 84% overall score, demonstrating strong technical and problem-solving skills.
  - **Hands-on Experience**: Worked on OOP concepts, database queries, and basic application development.

Projects

- **AI Document Intelligence Platform (RAG-based Chat System)**
  React, FastAPI, FAISS, Mixtral, LLaMA
  - **RAG Platform**: Built a production-ready Retrieval-Augmented Generation (RAG) system enabling context-aware document querying and conversational AI experiences.
  - **LLM Integration**: Integrated Mixtral and LLaMA models through OpenAI-compatible APIs to deliver accurate and scalable response generation.
  - **Backend Development**: Developed a FastAPI backend with streaming response support for low-latency, real-time user interactions.
  - **Semantic Search**: Implemented a FAISS-based retrieval pipeline to improve document relevance and reduce response latency.
  - **Frontend Development**: Built a React-based chat interface with ChatGPT-style conversations, multi-session support, and responsive user experience.

- **System Architecture**: Designed a modular embedding, retrieval, and generation pipeline while resolving CORS, streaming, and API compatibility challenges.

- **ShellMind – AI-Powered Linux Terminal Assistant**
  Python, NVIDIA NIM API, Typer, Rich
  - **Natural Language CLI**: Built an AI-powered Linux terminal assistant that converts natural language instructions into Linux commands and explains existing commands using NVIDIA NIM APIs.
  - **Safety Analysis**: Implemented real-time command risk assessment to detect potentially dangerous operations and recommend safer alternatives before execution.
  - **Package Distribution**: Developed a globally installable Python package using pip/pipx with modular architecture, environment-based configuration, and diagnostic tooling.
  - **Terminal Experience**: Created an enhanced terminal UI using Rich with structured outputs including command explanations, safety ratings, troubleshooting information, and diagnostics.

Technical Skills

- **Languages:** Python, C#, JavaScript, TypeScript, SQL
- **AI/ML:** LLMs, RAG, LangChain, AI Agents, Prompt Engineering, FAISS, Hugging Face, Ollama, NVIDIA NIM
- **Frontend:** React.js, HTML, CSS, JavaScript, TypeScript
- **Backend:** FastAPI, Django, Django REST Framework, .NET, REST APIs
- **Databases:** MySQL, Pinecone, FAISS
- **Cloud & DevOps:** Azure App Services, Azure Functions, Azure Key Vault, Git, Azure DevOps, CI/CD
- **Core Concepts:** DSA, OOP, System Design, SDLC, Agile

Certifications

- **Microsoft Azure Fundamentals (AZ-900)**:
- **Docker Training Course for the Absolute Beginner**:
- **Introduction to Containers, Kubernetes, and OpenShift V2 – IBM**:
- **Full Stack Development with Python & Django**:

Strengths

- : Dedicated Learner | Problem Solving | Strong Communication Skills | Lateral Thinking